

Troubleshooting in TCP/IP

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★ Packet Travelling

A data packet travels through a series of devices using routing protocols & IP addresses

Packet Creation : Sender Device - send request of data, broken down into packets
Encapsulation - each packet contains info. info ; Source IP address, destination IP address, data, TTL (time to live)

Local Area Network (LAN) : NIC sends the packets to local network over Wi-Fi or Ethernet or Switch or Router.

Router (First Hop) : Default Gateway
NAT (Network Address Translation) Private IP → Public IP

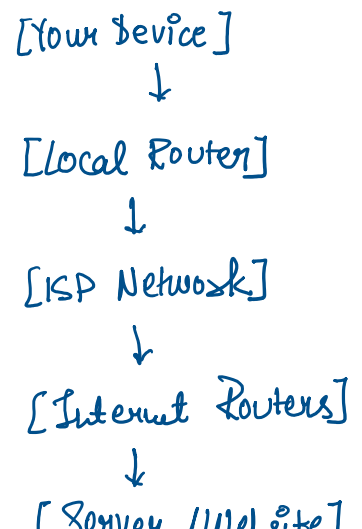
ISP Network : Router → ISP → best available path

Across the Internet (Intermediate Hops) : Multiple Routers
Best Path
Hops

Destination Network : Edge Router
Internal Network (Data Center)

Server (Destination)
Return Path

TTL (Time to Live) : value in a network packet that determines how or how many hops (routers) the packet is allowed to travel through before being discarded.



travel through before being discarded.

↓
[Server / Website]

Importance: Prevents infinite loops
Helps with Traceroute
Measures Distance (in hops)

64 : Linux/Unix

128 : Windows

255 : Maximum TTL

★ Troubleshooting

Process of identifying & resolving issues within a computer network

- Identifying Symptoms (recognizing problems)
- Gathering Information (collecting data - network, device, connections using tools)
- Analyzing Data (pinpoint the root cause)
- Isolating the problem (hardware, software or configuration)
- Developing a Plan (strategy)
- Implementing the Plan
- Verifying & Testing (confirming if the issue is resolved)

Troubleshooting Tools -

Allow to connect to the switch, run commands & manage configurations from a graphical interface.

→ Secure Shell (SSH) Client - Software Application that enables users to connect to a remote server or device over a network or internet connection using SSH protocol.

SSH Secure Shell Client (Windows)

→ Network Management Software - Cisco works, SolarWinds Network Performance Monitor (NPM)

→ Command-Line Interface (CLI) Tools - Cisco CLI, Arista CLI, Juniper CLI

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→ GUI based tools - Cisco DNA Center, Juniper Junos Space

★ Telnet

Network Protocol used for remote access to devices

Allows users to connect to a remote device & execute commands as if they were physically present at the device.



Security Limitations :

- Plain text transmission
- No encryption
- Weak authentication